

Prototype Overview: Automated Solar Panel Cleaning System

Team 0104A

1. Introduction

This document presents a comprehensive overview of the design, fabrication, integration, and verification of the Phase 3 Prototype System for the automated solar panel cleaning solution. The objective of this prototype was to validate the feasibility of a non-abrasive, film-based cleaning mechanism capable of removing dust while maintaining high light transmission and minimizing material consumption.

The final as-built prototype consists of a belt-driven transparent film system actuated by a stepper motor and controlled via a microcontroller with light-based activation. This document summarizes the system architecture, detailed design, key design decisions, integration challenges, and performance relative to the developed specifications, while referencing supporting artifacts across the Design Dossier.

2. System Architecture

2.1 Structural Subsystem

The structural frame was constructed from machine plywood (approximately 25 cm $\times$ 20 cm), selected for rapid fabrication and ease of modification (see 04.2 Build). Two parallel GT2 timing belts are routed along the frame and supported by front and rear idler pulleys mounted on wooden dowels.

A transparent polymer film spans the width of the system and is mechanically attached to the belts. The film travels across the acrylic panel surface, acting as a sacrificial cleaning interface. Unlike the conceptual design, the implemented system required the addition of a secondary film layer above the belts to ensure proper contact due to pulley height constraints.

Dust removal is achieved through a passive cleaning mechanism consisting of a microfiber cloth positioned along the return path of the film near the rear shaft. This location was selected to maximize cleaning effectiveness due to increased tension in the film during bending.

2.2 Electrical Design

The system is driven by a single NEMA 17 stepper motor connected to a shared wooden dowel shaft via a 3D-printed coupler. This configuration ensures synchronized motion across both belts (see 04.1-Pros and Cons Converging).

Motor control is achieved using an A4988 stepper driver, which interfaces with the Raspberry Pi Pico W microcontroller. The electrical subsystem was designed to minimize component count while ensuring sufficient torque to overcome frictional resistance (see 04.3 Integration).

2.3 Software Design

The control system utilizes a photoresistor (LDR) to detect reductions in light intensity caused by dust accumulation. Upon detecting a threshold drop in light transmission, the Pico W triggers a single motor rotation cycle.

This represents a significant deviation from the original periodic actuation concept, introducing an event-driven control strategy that reduces unnecessary energy consumption and aligns with system objectives for low intervention (see 04.1 Prototype Feedback Notes and Resulting Iteration and 04.4 VerificationProtocol+Results).

Figure 1 presents the system architecture of the as-built Prototype System, illustrating the interaction between the structural, electrical, and software subsystems.

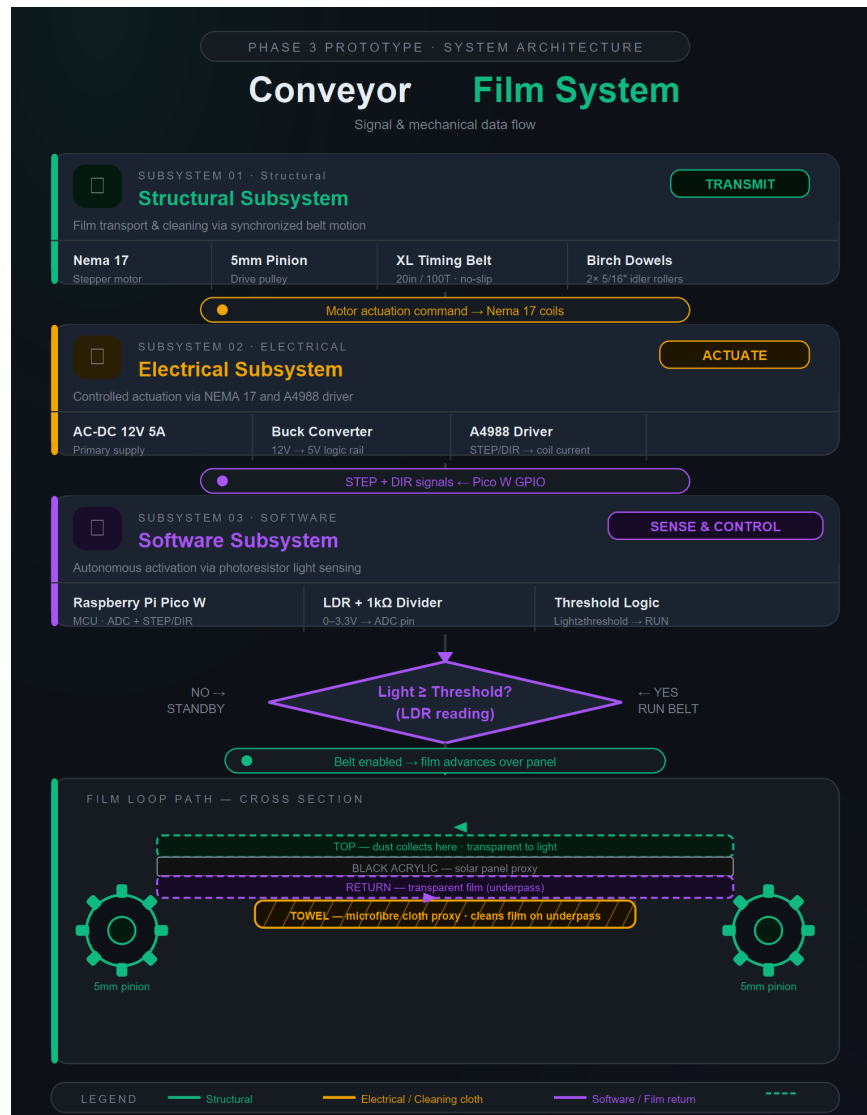


Figure 1: System architecture of the as-built prototype system.

The system integrates three primary subsystems:

Structural Subsystem: Responsible for film transport and cleaning action through synchronized belt motion.

Electrical Subsystem: Provides controlled actuation using a NEMA 17 stepper motor and A4988 driver.

Software Subsystem: Enables autonomous activation through a photoresistor-based light sensing mechanism.

Here is a more detailed overview of the 04.1-System Architecture

3. As-Built Prototype System

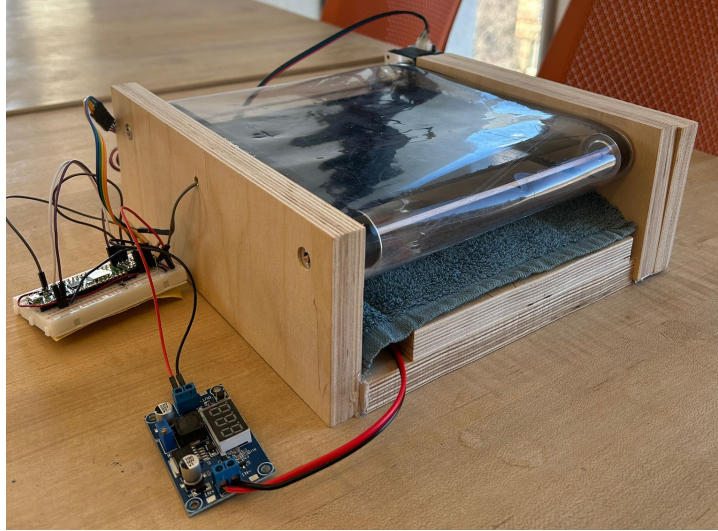


Figure 2: As-built prototype system.

The final prototype differs from the initial design concept in several critical ways, reflecting practical constraints and iterative refinement during Phase 3.

4. Design Process and Iteration

The design process followed a structured iterative workflow consisting of divergence, convergence, fabrication, and integration stages.

Initial concept generation and evaluation were conducted using brainstorming, SCAMPER, and Pugh analysis methods (see 04.1-PrototypeDesign). These were followed by a trade-off analysis to converge on final subsystem decisions, including motor selection, film material, and attachment methods.

Fabrication (see 04.2 Build) involved rapid prototyping using MyFab resources, enabling quick iteration on frame geometry and shaft configurations. Integration (see 04.3 Integration) revealed several mechanical and alignment challenges, which required refinement of shaft interfaces, belt alignment, and friction reduction strategies.

This iterative process ensured that the final prototype reflects both design intent and practical constraints encountered during implementation.

5. Key Design Decisions

Several critical design decisions significantly influenced the performance and feasibility of the prototype:

Single Motor Drive System: A single NEMA 17 motor driving a shared shaft was selected over dual motors to eliminate synchronization issues and ensure uniform belt motion.

Transparent Film Selection: A MyFab-sourced plastic sheet was chosen for its durability and flexibility, balancing optical clarity with mechanical robustness.

Hot Glue Film Attachment: Hot Glue was selected over Stitching methods for improved reliability during repeated cycles and if we made a mistake with stitching the film it would be very hard to remove.

Plywood Frame Construction: Laser-cut plywood enabled rapid iteration and sufficient structural integrity for the prototype scale.

Photoresistor-Based Control: Transitioning from periodic activation to light-triggered control improved energy efficiency and reduced unnecessary operation.

These decisions are supported by trade-off analysis and documented in 04.1-PrototypeDesign

6. Integration Challenges and Debugging

During integration, several significant challenges were encountered highlighted in :

Pulley Height Mismatch: The belt sat below the pulley edges, preventing proper film contact. This was resolved by layering additional material to raise the belt surface.

High Friction in Shaft Supports: Wood-on-wood contact created excessive resistance. This was mitigated by introducing a metal shaft coupler, reducing friction and improving rotational smoothness.

Dowel Sizing Constraints: Available dowel sizes did not match pulley interfaces, leading to alignment issues. The final solution utilized smaller dowels secured with screws to ensure proper alignment and consistent rotation.

These challenges and their resolutions are documented in detail in 04.3 Integration.

7. Performance Relative to Specifications

The Prototype System was evaluated against key structural, electrical, and software specifications (see 04-Specification and 04.4 VerificationProtocol+Results). All primary specifications were met under testing conditions.

Dust Removal (OM-1.1): Visual inspection across multiple trials confirmed that the microfiber cleaning mechanism consistently removed surface dust after a single rotation cycle, achieving a clear transition from a contaminated to a clean state.

Light Transmission (OM-1.2, OM-1.3): Inspection and testing verified that structural components obstructed less than 5% of the panel surface, and that light transmission improved after cleaning cycles, confirming removal of optically blocking debris.

Sustainability and Material Use (OM-2.1): The system operates without water or disposable components and maintained film integrity over repeated cycles, satisfying zero-consumable requirements.

Mechanical Safety (OM-2.6): Demonstration testing confirmed that sufficient clearance was maintained between the film and acrylic surface, with no observed scratching or deformation after operation.

Electrical Performance (OE-1.1, OE-1.4): The motor consistently achieved a full rotation under load without stalling, and cycle times remained within acceptable limits with smooth motion.

Software Control (OS-1.1, OS-1.5): The system successfully performed autonomous, single-rotation cleaning cycles triggered by light sensing, requiring no user intervention beyond power input.

Overall, the prototype satisfies all critical functional requirements and demonstrates feasibility of the proposed cleaning mechanism. Detailed procedures, raw data, and supporting media are provided in 04.4 Verification.

8. Contribution to Design Concept Understanding

The Prototype System provides strong validation of the proposed Design Concept and its alignment with the identified need of improving solar panel performance in dust-prone environments while maintaining low user burden. Can be found in 02-ESC204S 2026 BuildingSustainability Requirements

Validation of Need and Value Proposition: The prototype demonstrates that a passive, low-complexity system can effectively remove dust without requiring water, manual cleaning, or frequent maintenance. This directly supports the value proposition of enabling more sustainable building energy use through improved solar panel efficiency.

Alignment with Goal G-1 (Dust Removal and Energy Performance): The system successfully removes surface dust (OM-1.1) and improves light transmission (OM-1.3), validating the core assumption that a translating film mechanism can enhance solar panel performance. Additionally, minimal surface obstruction (OM-1.2) ensures that the system does not significantly reduce incoming sunlight, supporting the objective of maintaining or improving net energy output.

Alignment with Goal G-2 (Reliability and Low User Impact): The prototype confirms that the system can operate without consumables (OM-2.1), minimizing operational costs and maintenance requirements. Autonomous activation using a photoresistor (OS-1.5) eliminates the need for user intervention, while acceptable noise levels and smooth operation (OM-2.3, OE-1.4) contribute to usability in residential contexts.

Key Insights from Prototyping: The prototyping process revealed several important insights that refine the Design Concept:

- Film-based cleaning is effective, but performance depends strongly on maintaining consistent tension and alignment.
- Friction within the mechanical system significantly impacts motor performance and must be minimized through material and interface choices.
- Autonomous sensing is feasible with simple components, but requires careful calibration to avoid false triggers.
- Integration challenges (shaft alignment, belt height, material tolerances) are critical to system reliability and were not fully captured at the conceptual stage.

Implications for Future Design Iterations: While the prototype validates the core mechanism, scaling to real-world solar panels will require improvements in durability (e.g., UV-resistant films), more precise manufacturing tolerances, and optimized sensing thresholds. Nonetheless, the results confirm that the Design Concept is technically viable and aligned with the original Need, Goals, and Objectives defined in Phase 2.

9. Conclusion

The Phase 3 prototype successfully demonstrates the feasibility of an automated, non-abrasive solar panel cleaning system. Through iterative design, integration, and testing, the system validates key functional requirements while highlighting areas for future refinement. The results support the viability of the proposed design concept for further development and scaling.